

Analysis Airlines Project(2)

May 28, 2026

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

import a csv file

```
[2]: df=pd.read_csv('airlines_flights_data.csv')
df
```

```
[2]:
```

	index	airline	flight	source_city	departure_time	stops	\
0	0	SpiceJet	SG-8709	Delhi	Evening	zero	
1	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	
2	2	AirAsia	I5-764	Delhi	Early_Morning	zero	
3	3	Vistara	UK-995	Delhi	Morning	zero	
4	4	Vistara	UK-963	Delhi	Morning	zero	
...	...	...	...	...	...	...	
300148	300148	Vistara	UK-822	Chennai	Morning	one	
300149	300149	Vistara	UK-826	Chennai	Afternoon	one	
300150	300150	Vistara	UK-832	Chennai	Early_Morning	one	
300151	300151	Vistara	UK-828	Chennai	Early_Morning	one	
300152	300152	Vistara	UK-822	Chennai	Morning	one	

	arrival_time	destination_city	class	duration	days_left	price
0	Night	Mumbai	Economy	2.17	1	5953
1	Morning	Mumbai	Economy	2.33	1	5953
2	Early_Morning	Mumbai	Economy	2.17	1	5956
3	Afternoon	Mumbai	Economy	2.25	1	5955
4	Morning	Mumbai	Economy	2.33	1	5955
...	...	...	...	...	...	...
300148	Evening	Hyderabad	Business	10.08	49	69265
300149	Night	Hyderabad	Business	10.42	49	77105
300150	Night	Hyderabad	Business	13.83	49	79099
300151	Evening	Hyderabad	Business	10.00	49	81585
300152	Evening	Hyderabad	Business	10.08	49	81585

[300153 rows x 12 columns]

```
[3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300153 entries, 0 to 300152
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   index                 300153 non-null  int64
1   airline               300153 non-null  object
2   flight                300153 non-null  object
3   source_city           300153 non-null  object
4   departure_time        300153 non-null  object
5   stops                 300153 non-null  object
6   arrival_time          300153 non-null  object
7   destination_city      300153 non-null  object
8   class                 300153 non-null  object
9   duration               300153 non-null  float64
10  days_left             300153 non-null  int64
11  price                 300153 non-null  int64
dtypes: float64(1), int64(3), object(8)
memory usage: 27.5+ MB
```

```
[4]: df.columns
```

```
[4]: Index(['index', 'airline', 'flight', 'source_city', 'departure_time', 'stops',
        'arrival_time', 'destination_city', 'class', 'duration', 'days_left',
        'price'],
        dtype='object')
```

```
[5]: df.isnull().sum()
```

```
[5]: index                0
airline                0
flight                 0
source_city            0
departure_time         0
stops                  0
arrival_time           0
destination_city       0
class                  0
duration               0
days_left             0
price                  0
dtype: int64
```

data cleaning

```
[6]: # Remove unwanted Columns /as;- index columns is extra
df.drop(columns = 'index', inplace=True)
```

```
[7]: df
```

```
[7]:
```

	airline	flight	source_city	departure_time	stops	arrival_time	\
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	
4	Vistara	UK-963	Delhi	Morning	zero	Morning	
...	...	...	...	...	...	...	
300148	Vistara	UK-822	Chennai	Morning	one	Evening	
300149	Vistara	UK-826	Chennai	Afternoon	one	Night	
300150	Vistara	UK-832	Chennai	Early_Morning	one	Night	
300151	Vistara	UK-828	Chennai	Early_Morning	one	Evening	
300152	Vistara	UK-822	Chennai	Morning	one	Evening	

	destination_city	class	duration	days_left	price
0	Mumbai	Economy	2.17	1	5953
1	Mumbai	Economy	2.33	1	5953
2	Mumbai	Economy	2.17	1	5956
3	Mumbai	Economy	2.25	1	5955
4	Mumbai	Economy	2.33	1	5955
...	...	...	...	...	...
300148	Hyderabad	Business	10.08	49	69265
300149	Hyderabad	Business	10.42	49	77105
300150	Hyderabad	Business	13.83	49	79099
300151	Hyderabad	Business	10.00	49	81585
300152	Hyderabad	Business	10.08	49	81585

[300153 rows x 11 columns]

```
[8]: #statical summary of the data set
df.describe()
```

```
[8]:
```

	duration	days_left	price
count	300153.000000	300153.000000	300153.000000
mean	12.221021	26.004751	20889.660523
std	7.191997	13.561004	22697.767366
min	0.830000	1.000000	1105.000000
25%	6.830000	15.000000	4783.000000
50%	11.250000	26.000000	7425.000000
75%	16.170000	38.000000	42521.000000
max	49.830000	49.000000	123071.000000

```
[9]: # checking the flight durations was max
df[df['duration'] ==49.830000]
```

```
[9]:      airline flight source_city departure_time stops \
193889 Air_India AI-672 Chennai Evening two_or_more
194359 Air_India AI-672 Chennai Evening one

      arrival_time destination_city class duration days_left price
193889 Evening Bangalore Economy 49.83 2 23891
194359 Evening Bangalore Economy 49.83 9 17538
```

```
[10]: #checking the flight durations are Maximun
df[df['duration'] ==0.830000]
```

```
[10]:      airline flight source_city departure_time stops arrival_time \
115869 Indigo 6E-357 Bangalore Night zero Night
115943 Indigo 6E-357 Bangalore Night zero Night
116010 Indigo 6E-357 Bangalore Night zero Night
116081 Indigo 6E-357 Bangalore Night zero Night
116163 Indigo 6E-357 Bangalore Night zero Night
116236 Indigo 6E-357 Bangalore Night zero Night
116322 Indigo 6E-357 Bangalore Night zero Night
116411 Indigo 6E-357 Bangalore Night zero Night
116496 Indigo 6E-357 Bangalore Night zero Night
116656 Indigo 6E-357 Bangalore Night zero Night
116835 Indigo 6E-357 Bangalore Night zero Night
116924 Indigo 6E-357 Bangalore Night zero Night
117019 Indigo 6E-357 Bangalore Night zero Night
117101 Indigo 6E-357 Bangalore Night zero Night
117190 Indigo 6E-357 Bangalore Night zero Night
117274 Indigo 6E-357 Bangalore Night zero Night
117366 Indigo 6E-357 Bangalore Night zero Night
117461 Indigo 6E-357 Bangalore Night zero Night
117547 Indigo 6E-357 Bangalore Night zero Night
117643 Indigo 6E-357 Bangalore Night zero Night
117728 Indigo 6E-357 Bangalore Night zero Night
117817 Indigo 6E-357 Bangalore Night zero Night
117900 Indigo 6E-357 Bangalore Night zero Night
117995 Indigo 6E-357 Bangalore Night zero Night
118086 Indigo 6E-357 Bangalore Night zero Night
118173 Indigo 6E-357 Bangalore Night zero Night
118269 Indigo 6E-357 Bangalore Night zero Night
118355 Indigo 6E-357 Bangalore Night zero Night
118445 Indigo 6E-357 Bangalore Night zero Night
118528 Indigo 6E-357 Bangalore Night zero Night
118622 Indigo 6E-357 Bangalore Night zero Night
118712 Indigo 6E-357 Bangalore Night zero Night
```

118799	Indigo	6E-357	Bangalore	Night	zero	Night
118896	Indigo	6E-357	Bangalore	Night	zero	Night
118982	Indigo	6E-357	Bangalore	Night	zero	Night
119072	Indigo	6E-357	Bangalore	Night	zero	Night
119155	Indigo	6E-357	Bangalore	Night	zero	Night
197354	Indigo	6E-987	Chennai	Early_Morning	zero	Early_Morning
197355	Indigo	6E-477	Chennai	Early_Morning	zero	Morning
197356	Indigo	6E-6137	Chennai	Morning	zero	Morning
197445	Indigo	6E-987	Chennai	Early_Morning	zero	Early_Morning
197446	Indigo	6E-477	Chennai	Early_Morning	zero	Morning
197447	Indigo	6E-6137	Chennai	Morning	zero	Morning
197537	Indigo	6E-987	Chennai	Early_Morning	zero	Early_Morning
197538	Indigo	6E-477	Chennai	Early_Morning	zero	Morning
197539	Indigo	6E-6137	Chennai	Morning	zero	Morning
197626	Indigo	6E-987	Chennai	Early_Morning	zero	Early_Morning
197627	Indigo	6E-477	Chennai	Early_Morning	zero	Morning
197628	Indigo	6E-6137	Chennai	Morning	zero	Morning
197712	Indigo	6E-987	Chennai	Early_Morning	zero	Early_Morning
197713	Indigo	6E-477	Chennai	Early_Morning	zero	Morning
197724	Indigo	6E-6137	Chennai	Morning	zero	Morning

	destination_city	class	duration	days_left	price
115869	Chennai	Economy	0.83	6	3498
115943	Chennai	Economy	0.83	7	3498
116010	Chennai	Economy	0.83	8	3498
116081	Chennai	Economy	0.83	9	3498
116163	Chennai	Economy	0.83	10	3498
116236	Chennai	Economy	0.83	11	3498
116322	Chennai	Economy	0.83	12	3498
116411	Chennai	Economy	0.83	13	3498
116496	Chennai	Economy	0.83	14	3498
116656	Chennai	Economy	0.83	16	1924
116835	Chennai	Economy	0.83	18	1924
116924	Chennai	Economy	0.83	19	1924
117019	Chennai	Economy	0.83	20	1924
117101	Chennai	Economy	0.83	21	1924
117190	Chennai	Economy	0.83	22	1924
117274	Chennai	Economy	0.83	23	1924
117366	Chennai	Economy	0.83	24	1924
117461	Chennai	Economy	0.83	25	1924
117547	Chennai	Economy	0.83	26	1924
117643	Chennai	Economy	0.83	27	1924
117728	Chennai	Economy	0.83	28	1924
117817	Chennai	Economy	0.83	29	1924
117900	Chennai	Economy	0.83	30	1924
117995	Chennai	Economy	0.83	31	1604
118086	Chennai	Economy	0.83	32	1604

118173	Chennai	Economy	0.83	33	1604
118269	Chennai	Economy	0.83	34	1604
118355	Chennai	Economy	0.83	35	1604
118445	Chennai	Economy	0.83	36	1604
118528	Chennai	Economy	0.83	37	1604
118622	Chennai	Economy	0.83	38	1604
118712	Chennai	Economy	0.83	39	1604
118799	Chennai	Economy	0.83	40	1604
118896	Chennai	Economy	0.83	41	1604
118982	Chennai	Economy	0.83	42	1604
119072	Chennai	Economy	0.83	43	1604
119155	Chennai	Economy	0.83	44	1604
197354	Bangalore	Economy	0.83	45	1443
197355	Bangalore	Economy	0.83	45	1443
197356	Bangalore	Economy	0.83	45	1443
197445	Bangalore	Economy	0.83	46	1443
197446	Bangalore	Economy	0.83	46	1443
197447	Bangalore	Economy	0.83	46	1443
197537	Bangalore	Economy	0.83	47	1443
197538	Bangalore	Economy	0.83	47	1443
197539	Bangalore	Economy	0.83	47	1443
197626	Bangalore	Economy	0.83	48	1443
197627	Bangalore	Economy	0.83	48	1443
197628	Bangalore	Economy	0.83	48	1443
197712	Bangalore	Economy	0.83	49	1443
197713	Bangalore	Economy	0.83	49	1443
197724	Bangalore	Economy	0.83	49	1549

```
[11]: #checking the Maximum flight price which
df[df['price']==123071.000000]
```

```
[11]:      airline  flight source_city departure_time stops arrival_time \
261377  Vistara  UK-772      Kolkata      Morning     one      Night

      destination_city      class  duration  days_left  price
261377             Delhi  Business      13.5          3  123071
```

```
[12]: #checking the Minimum flight price which
df[df['price']==1105.000000]
```

```
[12]:      airline  flight source_city departure_time stops arrival_time \
203807  AirAsia  I5-517      Chennai      Morning     zero      Morning
203808  GO_FIRST  G8-505      Chennai      Evening     zero      Evening
203908  AirAsia  I5-517      Chennai      Morning     zero      Morning
203909  GO_FIRST  G8-505      Chennai      Evening     zero      Evening
204003  AirAsia  I5-517      Chennai      Morning     zero      Morning
...      ...      ...      ...      ...      ...      ...
```

206601	Indigo	6E-7261	Chennai	Morning	one	Evening
206602	Indigo	6E-611	Chennai	Evening	one	Late_Night
206603	Indigo	6E-581	Chennai	Morning	one	Evening
206604	Indigo	6E-7127	Chennai	Afternoon	one	Night
206605	Indigo	6E-7261	Chennai	Morning	one	Night

	destination_city	class	duration	days_left	price
203807	Hyderabad	Economy	1.17	16	1105
203808	Hyderabad	Economy	1.25	16	1105
203908	Hyderabad	Economy	1.17	17	1105
203909	Hyderabad	Economy	1.25	17	1105
204003	Hyderabad	Economy	1.17	18	1105
...	...	...	...	...	...
206601	Hyderabad	Economy	7.92	49	1105
206602	Hyderabad	Economy	8.25	49	1105
206603	Hyderabad	Economy	9.17	49	1105
206604	Hyderabad	Economy	9.50	49	1105
206605	Hyderabad	Economy	10.08	49	1105

[300 rows x 11 columns]

```
[13]: # checking the missing value in data set/any value
df.isnull().sum()
```

```
[13]: airline      0
flight         0
source_city    0
departure_time 0
stops          0
arrival_time   0
destination_city 0
class          0
duration       0
days_left     0
price          0
dtype: int64
```

1 Q.1:- What is the airlines in the dataset,accompanied by their fequenice?

```
[14]: df.head()
```

```
[14]:   airline  flight source_city departure_time stops arrival_time \
0  SpiceJet  SG-8709      Delhi      Evening    zero      Night
1  SpiceJet  SG-8157      Delhi  Early_Morning    zero      Morning
2  AirAsia   I5-764      Delhi  Early_Morning    zero  Early_Morning
```

3	Vistara	UK-995	Delhi	Morning	zero	Afternoon
4	Vistara	UK-963	Delhi	Morning	zero	Morning

	destination_city	class	duration	days_left	price
0	Mumbai	Economy	2.17	1	5953
1	Mumbai	Economy	2.33	1	5953
2	Mumbai	Economy	2.17	1	5956
3	Mumbai	Economy	2.25	1	5955
4	Mumbai	Economy	2.33	1	5955

```
[15]: # Checking how many Airlines are in the dataset& checking unique value
df['airline'].nunique()
```

```
[15]: 6
```

```
[16]: # Showing the names of the Airlines are in the dataset
df['airline'].unique()
```

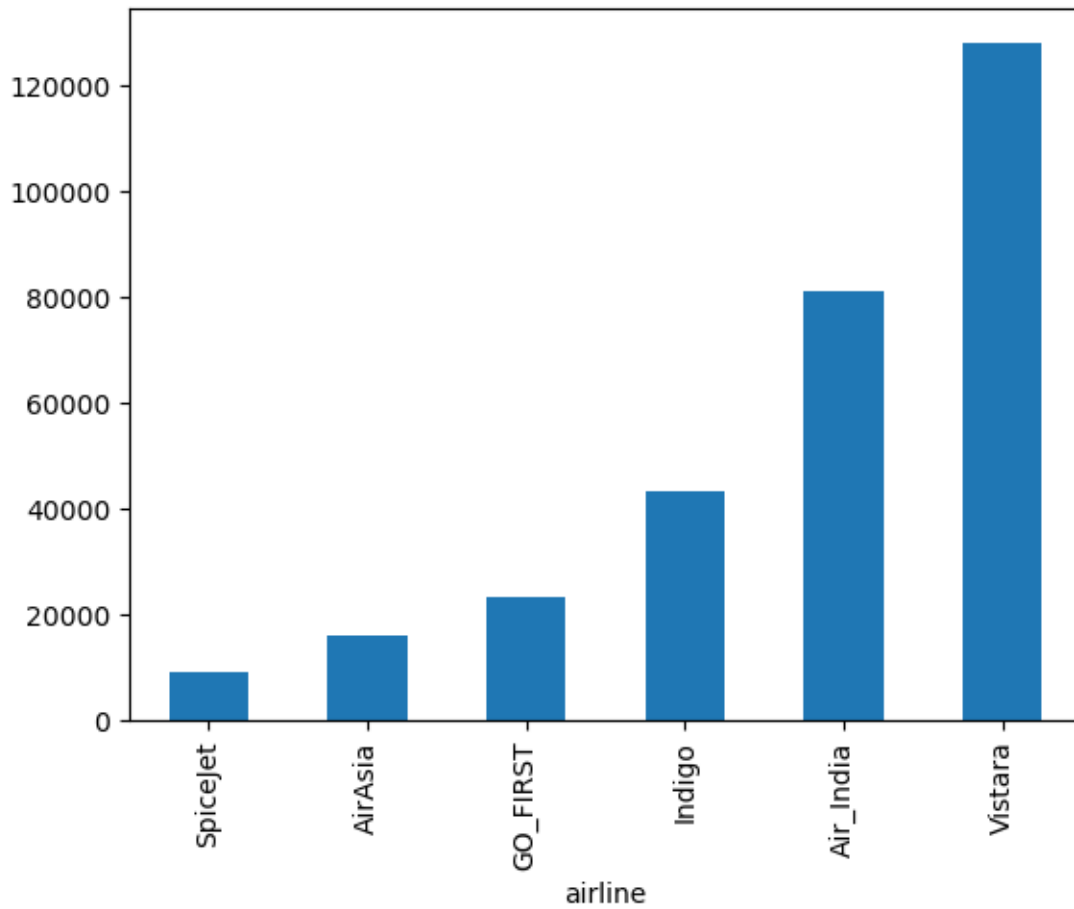
```
[16]: array(['SpiceJet', 'AirAsia', 'Vistara', 'GO_FIRST', 'Indigo',
        'Air_India'], dtype=object)
```

```
[17]: # showing all the Airlines with their frequncies
df['airline'].value_counts()
```

```
[17]: airline
Vistara      127859
Air_India    80892
Indigo       43120
GO_FIRST     23173
AirAsia      16098
SpiceJet     9011
Name: count, dtype: int64
```

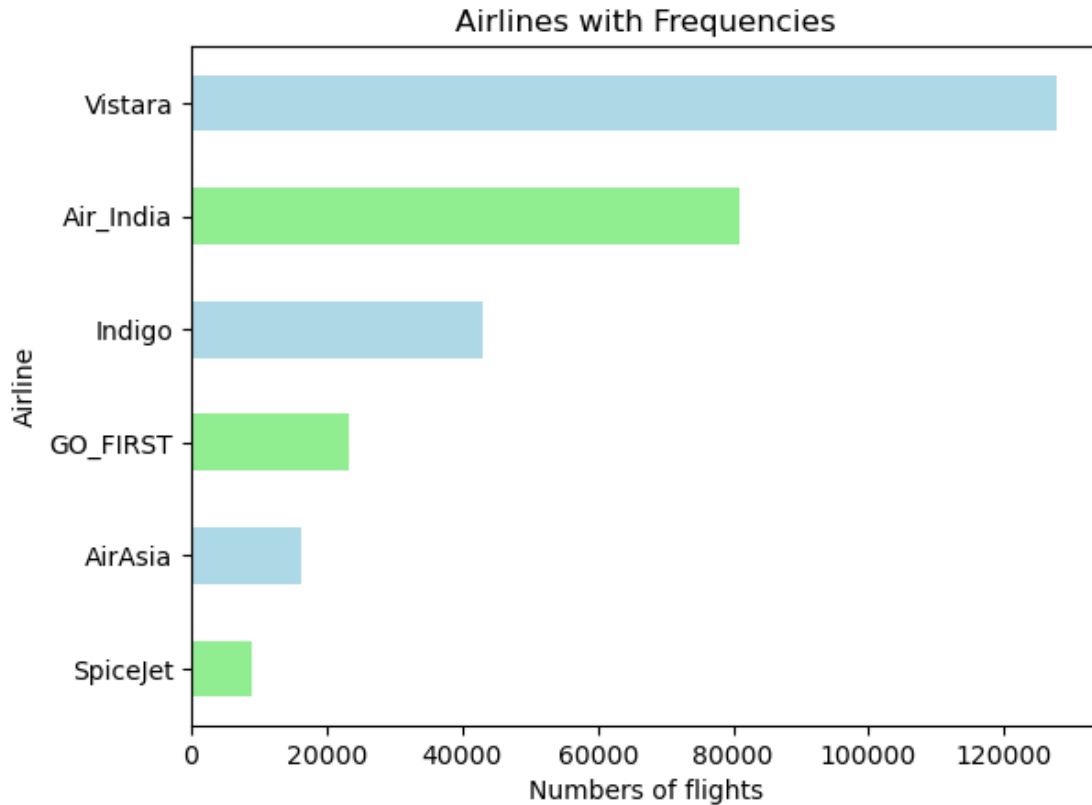
```
[18]: # showing all the Airlines with their Numbers of flights in Bar graph

df['airline'].value_counts(ascending=True).plot.bar()
plt.show()
```

```
[19]: # showing all the Airlines with their Numbers of flights in horizontal Bar
      ↳ graph same as upper but graph is horizontal
```

```
df['airline'].value_counts(ascending=True).plot.
  ↳ barh(color=['lightgreen', 'lightblue'])
plt.title("Airlines with Frequencies")
plt.xlabel("Numbers of flights")
plt.ylabel("Airline")
plt.show()
```



Q2:- show Bar Graphs representing the Departure time & Arrival time

```
[20]: df.head()
```

```
[20]:
```

	airline	flight	source_city	departure_time	stops	arrival_time	\
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	
4	Vistara	UK-963	Delhi	Morning	zero	Morning	

	destination_city	class	duration	days_left	price
0	Mumbai	Economy	2.17	1	5953
1	Mumbai	Economy	2.33	1	5953
2	Mumbai	Economy	2.17	1	5956
3	Mumbai	Economy	2.25	1	5955
4	Mumbai	Economy	2.33	1	5955

```
[21]: # showing the Departure time for the flights
df['departure_time'].value_counts()
```

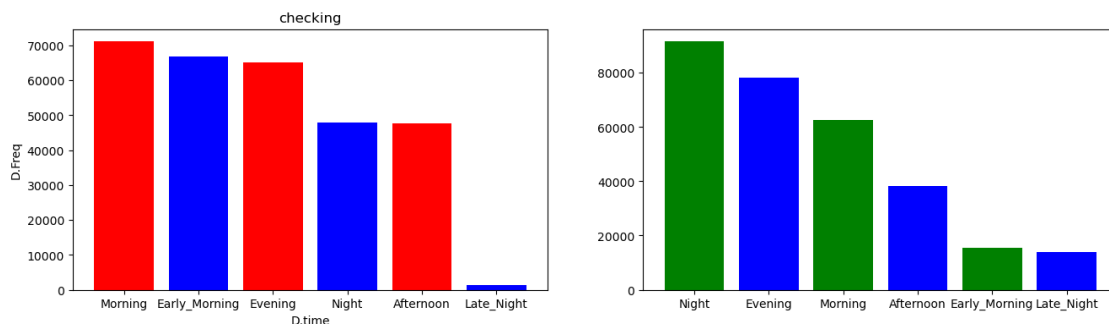
```
[21]: departure_time
      Morning      71146
      Early_Morning 66790
      Evening      65102
      Night        48015
      Afternoon    47794
      Late_Night   1306
      Name: count, dtype: int64
```

```
[22]: # showing the Arrival time for the flights
      df['arrival_time'].value_counts()
```

```
[22]: arrival_time
      Night      91538
      Evening   78323
      Morning   62735
      Afternoon 38139
      Early_Morning 15417
      Late_Night 14001
      Name: count, dtype: int64
```

```
[23]: # showing the Departure time & arrival time for the flights with their counts
      plt.figure(figsize=(16,4))
      plt.subplot(1,2,1)
      plt.bar(df['departure_time'].value_counts().index,df['departure_time'].
      ↪value_counts().values,color=['r','b'])
      plt.title("checking")
      plt.ylabel("D.Freq")
      plt.xlabel("D.time")

      plt.subplot(1,2,2)
      plt.bar(df['arrival_time'].value_counts().index,
      df['arrival_time'].value_counts().values,color=['g','b'])
      plt.show()
```



Q3:- show the Bar graph representing the source City & Destination City

```
[24]: df.head()
```

```
[24]:   airline  flight source_city departure_time stops  arrival_time \
0  SpiceJet  SG-8709      Delhi      Evening    zero      Night
1  SpiceJet  SG-8157      Delhi  Early_Morning    zero      Morning
2  AirAsia   I5-764      Delhi  Early_Morning    zero  Early_Morning
3  Vistara   UK-995      Delhi      Morning    zero      Afternoon
4  Vistara   UK-963      Delhi      Morning    zero      Morning

   destination_city  class  duration  days_left  price
0          Mumbai  Economy      2.17         1   5953
1          Mumbai  Economy      2.33         1   5953
2          Mumbai  Economy      2.17         1   5956
3          Mumbai  Economy      2.25         1   5955
4          Mumbai  Economy      2.33         1   5955
```

```
[25]: df['destination_city'].value_counts()
```

```
[25]: destination_city
Mumbai      59097
Delhi       57360
Bangalore   51068
Kolkata     49534
Hyderabad   42726
Chennai     40368
Name: count, dtype: int64
```

```
[26]: df['source_city'].value_counts()
```

```
[26]: source_city
Delhi       61343
Mumbai      60896
Bangalore   52061
Kolkata     46347
Hyderabad   40806
Chennai     38700
Name: count, dtype: int64
```

```
[27]: # showing the flights ,sources city and destination city and their counts and
      plot & subbar graphs
plt.figure(figsize=(16,4))
plt.subplot(1,2,1)

plt.barh(df['source_city'].value_counts().index,
         df['source_city'].value_counts().values,
         color=['m','g'])
```

```
plt.title("source Cities with No of flights")
plt.xlabel("NO of flights")
plt.ylabel("Name of cities")

# plotting a second subbur graphs
plt.barh(
    df['destination_city'].value_counts().index,
    df['destination_city'].value_counts().values,
    color=['g','b']
plt.title("sources cities with no flights")
plt.xlabel("no of flights")
plt.ylabel("Cities" )
plt.show()
```

Cell In[27], line 16

```
color=['g','b']
    ^
```

SyntaxError: invalid syntax. Perhaps you forgot a comma?

```
[ ]: # try with diffrent color
plt.figure(figsize=(16,4))

plt.subplot(1,2,1)

plt.barh(
    df['source_city'].value_counts().index,
    df['source_city'].value_counts().values,
    color=['r','b']
)

plt.show()
```

Q4:- Does price varies with airline

```
[ ]: df.head()
```

```
[ ]: # grouping the airline and checkig their mean price
df.groupby('airline')['price'].mean()
```

```
[ ]: df.describe()
```

```
[ ]: #Drawing a Categorical Plot showing the Mean Ticket Price for each Airline

sns.catplot(x = 'airline', y = 'price', kind = 'bar', palette = 'rocket', data= df)

plt.show()
```

2 Q4- Does ticket Price change based on the Departure time and Arrival time

```
[ ]: df.head(10)
```

```
[ ]: #checking the Mean Ticket Price based on The Departure Time
df.groupby('departure_time')['price'].mean()
```

```
[ ]: #checking the Mean Ticket Price based on The Arival Time
df.groupby('arrival_time')['price'].mean()
```

```
[ ]: sns.catplot(x = 'departure_time', y = 'price', kind = 'bar', data= df)
```

```
[ ]: sns.catplot(x = 'arrival_time', y = 'price', kind = 'bar', data= df)
```

```
[ ]: # plotting a Realtional Grpahs
sns.relplot( x = 'arrival_time', y= 'price', data = df, col= 'departure_time', kind = 'line')
↪ 'line')
```

3 Q5:- How the price change with change in source and destination

```
[ ]: # checking the Mean ticket price for each source city
df.groupby('source_city')['price'].mean()
```

```
[ ]: # checking the Mean ticket price for each Destination city
df.groupby('destination_city')['price'].mean()
```

```
[ ]: # plotting a Realtional Grpahs
sns.relplot( x = 'destination_city', y= 'price', data = df, col= "source_city", kind = 'line')
↪ 'line')
```

4 Q6 :-How is the price affected when ticket are bought in just 1 or 2 days before depature?

```
[ ]: df.head()

[ ]: # check The Unique Count/ yani Number
df['days_left'].nunique()

[ ]: df['days_left'].unique()

[ ]: #Checking thr Mean Ticket Price For the Different days left
df.groupby('days_left')['price'].mean()

[ ]: sns.relplot(y='price',x='days_left',kind = 'line' ,data = df)
plt.show()
```

5 Q7 how does the ticket price vary B/w Economy and bussines class

```
[ ]: df.head()

[ ]: df['class'].nunique()

[ ]: df['class'].unique()

[33]: x = df[df['class'] == 'Economy']

x
```

```
[33]:
```

	airline	flight	source_city	departure_time	stops	arrival_time \
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon
4	Vistara	UK-963	Delhi	Morning	zero	Morning
...	...	...	...	...	...	...
206661	Vistara	UK-832	Chennai	Early_Morning	one	Night
206662	Vistara	UK-832	Chennai	Early_Morning	one	Night
206663	Vistara	UK-826	Chennai	Afternoon	one	Morning
206664	Vistara	UK-822	Chennai	Morning	one	Morning
206665	Vistara	UK-824	Chennai	Night	one	Night

	destination_city	class	duration	days_left	price
0	Mumbai	Economy	2.17	1	5953
1	Mumbai	Economy	2.33	1	5953
2	Mumbai	Economy	2.17	1	5956

3	Mumbai	Economy	2.25	1	5955
4	Mumbai	Economy	2.33	1	5955
...	...	...	...	...	...
206661	Hyderabad	Economy	13.83	49	7697
206662	Hyderabad	Economy	13.83	49	7709
206663	Hyderabad	Economy	20.58	49	8640
206664	Hyderabad	Economy	23.33	49	8640
206665	Hyderabad	Economy	24.42	49	8640

[206666 rows x 11 columns]

```
[34]: # checking the mean price of economy column
x.price.mean()
```

```
[34]: np.float64(6572.342383362527)
```

```
[44]: y = df[df['class'] == 'Business']

y
```

```
[44]:
```

	airline	flight	source_city	departure_time	stops	arrival_time	\
206666	Air_India	AI-868	Delhi	Evening	zero	Evening	
206667	Air_India	AI-624	Delhi	Evening	zero	Night	
206668	Air_India	AI-531	Delhi	Evening	one	Night	
206669	Air_India	AI-839	Delhi	Night	one	Night	
206670	Air_India	AI-544	Delhi	Evening	one	Night	
...	...	...	...	...	...	...	
300148	Vistara	UK-822	Chennai	Morning	one	Evening	
300149	Vistara	UK-826	Chennai	Afternoon	one	Night	
300150	Vistara	UK-832	Chennai	Early_Morning	one	Night	
300151	Vistara	UK-828	Chennai	Early_Morning	one	Evening	
300152	Vistara	UK-822	Chennai	Morning	one	Evening	

	destination_city	class	duration	days_left	price
206666	Mumbai	Business	2.00	1	25612
206667	Mumbai	Business	2.25	1	25612
206668	Mumbai	Business	24.75	1	42220
206669	Mumbai	Business	26.50	1	44450
206670	Mumbai	Business	6.67	1	46690
...	...	...	...	...	...
300148	Hyderabad	Business	10.08	49	69265
300149	Hyderabad	Business	10.42	49	77105
300150	Hyderabad	Business	13.83	49	79099
300151	Hyderabad	Business	10.00	49	81585
300152	Hyderabad	Business	10.08	49	81585

[93487 rows x 11 columns]


```
[45]: y.price .mean()
```

```
[45]: np.float64(52540.08112357868)
```

6 Q8:- what will be the AVg price of vistra airline for a flight from DELhi to hyderabad in Business class?

```
[48]: df.head()
```

```
[48]:
```

	airline	flight	source_city	departure_time	stops	arrival_time	\
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	
4	Vistara	UK-963	Delhi	Morning	zero	Morning	

	destination_city	class	duration	days_left	price
0	Mumbai	Economy	2.17	1	5953
1	Mumbai	Economy	2.33	1	5953
2	Mumbai	Economy	2.17	1	5956
3	Mumbai	Economy	2.25	1	5955
4	Mumbai	Economy	2.33	1	5955

```
[62]: new_data= df[(df['airline'] == 'Vistara') &
(df['source_city'] == 'Delhi') &
( df[ 'destination_city']=='Hyderabad')&
( df['class']== 'Business')]
new_data
```

```
[62]:
```

	airline	flight	source_city	departure_time	stops	arrival_time	\
219123	Vistara	UK-871	Delhi	Night	zero	Night	
219124	Vistara	UK-879	Delhi	Evening	zero	Evening	
219129	Vistara	UK-955	Delhi	Evening	one	Night	
219130	Vistara	UK-955	Delhi	Evening	one	Afternoon	
219131	Vistara	UK-985	Delhi	Evening	one	Night	
...	...	...	...	...	...	...	
221863	Vistara	UK-963	Delhi	Morning	one	Early_Morning	
221864	Vistara	UK-985	Delhi	Evening	one	Early_Morning	
221865	Vistara	UK-985	Delhi	Evening	one	Afternoon	
221866	Vistara	UK-955	Delhi	Evening	one	Early_Morning	
221867	Vistara	UK-955	Delhi	Evening	one	Afternoon	

	destination_city	class	duration	days_left	price
219123	Hyderabad	Business	2.33	1	30630
219124	Hyderabad	Business	2.25	1	38470
219129	Hyderabad	Business	27.17	1	63513

219130	Hyderabad	Business	18.50	1	65764
219131	Hyderabad	Business	25.08	1	69113
...	...	...	...	...	...
221863	Hyderabad	Business	23.00	49	53937
221864	Hyderabad	Business	12.00	49	59537
221865	Hyderabad	Business	16.42	49	59537
221866	Hyderabad	Business	14.08	49	61889
221867	Hyderabad	Business	18.50	49	61889

[1660 rows x 11 columns]

```
[63]: new_data['price'].mean()
```

```
[63]: np.float64(47939.840361445786)
```

```
[ ]:
```